Classical Complexity and Curvature-Bounded Physical Computation

A Critical Reassessment of the Proposed Physical P versus NP Separation

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Original research note: December 1, 2025
Revised mathematical audit: October 27, 2026

Abstract

This technical revision distinguishes the mathematically well-defined classical P versus NP question from questions of realizability and resource usage in Wave Curvature Computation (WCC). The discrete finite-state, local-update model can represent polynomial-time Turing computations under uniform encoding, decoding, and transition assumptions. That equivalence does not prove that each symbolic computation has a polynomial-curvature physical realization. Several claims in the original December 2025 technical note required correction: physical distinguishability alone does not force positive Laplacian-curvature cost; finite first-derivative energy and locality do not imply a universal polynomial bound on second-derivative curvature; expander graphs do not, without a separate geometric embedding theorem, imply exponential interface area; and the total influence of any n-bit Boolean function is at most n. We establish exact counterexamples and corrected conditional estimates, clarify physical P and NP witness admissibility, and isolate explicit proof obligations for a possible restricted-model separation. No unconditional separation of classical P and NP, nor a proved separation of the curvature-bounded physical classes, is claimed here.

New decoder-uniformity obstruction (November 2, 2026). A fixed, language-independent sign map encodes every n-bit string into n real amplitudes ± 1, with squared mass n, and the sign receiver recovers the string under per-coordinate additive errors with magnitude strictly below one. Composing this receiver with an arbitrary predicate would decide that predicate without charging the final decoding step. Such a decoder need not be computable, let alone efficient. Consequently, 2^n recoverable labels, finite amplitude cost, and a uniform encoding cannot establish an unconditional lower bound for decision problems without bounding the decoder's complexity. This counterexample does not establish an efficient SAT algorithm. The physical continuum-to-computation bridge and universal curvature-versus-decision lower bound remain open.

1 Historical Context: The Divergence of Logic and Physics

The classical P versus NP question concerns language classes defined by uniform Turing-machine running time and polynomial verification; the question is mathematically well posed independent of a physical implementation [9, 11]. Turing machines are abstract algorithms and do not assert that unbounded physical memory is available at zero energetic cost. Whether a concrete physical computer efficiently realizes a particular algorithm is an additional question. Locality, thermodynamic dissipation, finite precision, field regularity, and resource budgets can restrict a physical substrate without changing the definition of classical P or NP.

WCT motivates such a resource-sensitive investigation. We distinguish three notions throughout: classical classes P, NP; uniformly specified discrete WCC classes $P^{\mathrm{disc}}_{\mathrm{WCC}}, NP^{\mathrm{disc}}_{\mathrm{WCC}}$; and

physical, curvature-rail classes $P^{\text{phys}}_{\text{WCC}}, NP^{\text{phys}}_{\text{WCC}}$. A physical-vs-classical comparison must specify a uniform implementation map and its costs before inferring equality or separation of these classes.

2 The Physical Distinguishability Obstruction

2.1 Physical Encoding of Logical Bits

Let $\Omega \subset \mathbb{R}^3$ be a bounded Lipschitz domain, with periodic or Neumann boundary conditions allowed. Two field encodings $\psi_0, \psi_1 \in H^2(\Omega)$ can be distinct in the H^1 norm without having different Laplacians. We therefore distinguish *symbolic separation* from *curvature cost.*

Definition 2.1 (Regularized curvature cost)**.** *For $\varepsilon > 0$ and $\psi \in H^2(\Omega; \mathbb{C})$ define the complex-safe curvature operator*

$$\Theta_\varepsilon[\psi] = -\frac{\Delta\psi\,\overline{\psi}}{|\psi|^2 + \varepsilon^2}, \qquad C_{\Theta,\varepsilon}[\psi] = \int_\Omega |\Theta_\varepsilon[\psi]|^2\, dx.$$

For a pair of encodings, a proposed separation cost is $C_{\Theta,\varepsilon}[\psi_1 - \psi_0]$. This particular functional measures second-derivative structure, not every possible resource cost of a bit.

Proposition 2.2 (Counterexample to a universal positive separation cost)**.** *Distinct H^1 states need not have positive curvature-separation cost.*

Proof. Set $\psi_0 \equiv 0$ and $\psi_1 \equiv 1$ on Ω. Then $\|\psi_1 - \psi_0\|^2_{H^1} = |\Omega| > 0$ while $\Delta(\psi_1 - \psi_0) = 0$. Hence

$$C_{\Theta,\varepsilon}[\psi_1 - \psi_0] = 0.$$

The counterexample obeys the stated periodic or Neumann conditions. □

Remark 2.3 (WIN-12)**.** *The original Lemma 2.2 inferred nonzero Laplacian from nonzero H^1 distance, which is false. An experimentally grounded lower bound for a specific device would require extra boundary conditions, state constraints, and a coercivity inequality; distinguishability alone is insufficient.*

2.2 Curvature Cost of an n-Bit Configuration

A lower bound $c_0 n$ for storing n bits may be established within a specified *independent-cell encoding* if every occupied cell contributes at least $c_0 > 0$ and the chosen functional is additive across the disjoint supports. Neither cell independence nor positive c_0 follows from the previous definition. In particular, an n-bit string is not required to simultaneously realize the 2^n possible assignments.

Proposition 2.4 (Conditional additive storage estimate)**.** *If $\Omega = \bigsqcup_{j=1}^n \Omega_j$ and a chosen non-negative storage functional E_{store} is additive on the disjoint cells with $E_{\text{store}}(\psi_{b_j}|_{\Omega_j}) \geq c_0$ for each j, then $E_{\text{store}}(\psi_w) \geq nc_0$.*

Proof. Sum the n cellwise inequalities. No such conclusion is asserted for $C_{\Theta,\varepsilon}$ without proving the premises. □

2.3 Classical Turing Machines and Physical Implementations

The Turing model accounts for operations through a formal step count, not through a PDE-based curvature-energy functional. Its omission of an explicit energy budget does not imply a theorem that every physical implementation must consume zero energy or that a polynomial-overhead physical simulation is impossible. The physical cost of symbols, memory, and erasure is an independent quantitative engineering question [3, 4].

2.4 Physical Complexity Classes

For a fixed, uniform physical model $\mathfrak{M}$ and a stated polynomial resource rail R, define $P^{\mathrm{phys}}_{\mathfrak{M},R}$ by uniformly specified deterministic deciders that meet the rail on every input. Define $NP^{\mathrm{phys}}_{\mathfrak{M},R}$ with polynomial-length, physically admissible witness encodings and deterministic verifiers subject to the same rail. The physical classes are machine-relative; we do not identify them with classical classes without separately proving the necessary simulation theorems.

3 Classical Computation and Physical Resource Models

The classical formalism is mathematically defined even if no hardware can realize an unbounded tape for all input sizes at once. Complexity theory uses a *family* of finite computations, one for each input length. Physical realizability concerns whether a single uniform device family has allowed costs, scaling, and error control. The extended Church–Turing principle is a statement about efficient simulation of physical models, not a prerequisite for formulating classical P versus NP.

Remark 3.1. *A distinction between two physical encoding conventions cannot by itself refute a theorem about abstract languages. Conversely, a formal equivalence of symbolic update systems does not guarantee faithful realization by a specified nonlinear PDE under fixed energy and stability rails.*

4 The WCC Model and Curvature Axioms

Definition 4.1 (Polynomial curvature rail)**.** *For a uniform physical device family indexed by input length n, a trajectory $\psi_{n,x}(t)$ respects a polynomial curvature rail if there exists one polynomial p such that, for every input x of length n and all times $0 \le t \le T(n)$,*

$$\sup_{|x|=n} \sup_{0\le t\le T(n)} C_{\Theta,\varepsilon}[\psi_{n,x}(t)] \le p(n).$$

A requirement of finite *curvature at each n is strictly weaker.*

Lemma 4.2 (A valid conditional curvature bound)**.** *For every $\psi \in H^2(\Omega;\mathbb{C})$ and $\varepsilon > 0$,*

$$C_{\Theta,\varepsilon}[\psi] \le \frac{1}{4\varepsilon^2}\|\Delta\psi\|^2_{L^2(\Omega)}.$$

Proof. For $r \ge 0$, $r/(r^2+\varepsilon^2) \le 1/(2\varepsilon)$. Apply this pointwise to $r = |\psi|$ and integrate the square. □

Remark 4.3 (WIN-13)**.** *The original operator $-\Delta\psi/(\psi+\varepsilon)$ does not have the claimed uniform denominator bound $|\psi+\varepsilon| \ge \varepsilon$: $\psi = -\varepsilon$ is an immediate counterexample. One may instead assume an explicit gap $|\psi+\varepsilon| \ge \delta > 0$ or use the complex-safe regularizer of Definition 2.1. All estimates in this revision use the latter.*

5 Discrete and Physical WCC Semantics

5.1 Discrete WCC Model

A discrete WCC machine uses a finite alphabet S, a fixed-radius local update rule, polynomially many lattice cells, a uniform family of lattices, and polynomial-time constructible encoders and decoders. Machine descriptions must not contain nonuniform advice or a hard language in the initialization map.

Theorem 5.1 (Discrete Turing equivalence under uniformity)**.** *For the uniform finite-state, fixed-radius WCC model defined above, with polynomial configuration size and polynomial rounds,*

$$P_{\mathrm{WCC}}^{\mathrm{disc}} = P, \qquad NP_{\mathrm{WCC}}^{\mathrm{disc}} = NP.$$

Proof. A Turing machine sequentially updates all polynomially many cells in polynomial time per global WCC step. Conversely, standard finite-state local tape-and-head encoding simulates each step of a fixed polynomial-time Turing machine by a constant number of local rounds on a polynomial-size lattice. The two simulations preserve polynomial witness length and verifier runtime. Polynomial-time uniform encoding, decoding, and lattice generation exclude hidden oracle computations. □

5.2 Physical WCC Model and Witness Admissibility

The physical model additionally restricts allowable embeddings, precision, curvature-energy budgets, and dynamics. Therefore the discrete theorem alone supplies no equality $P_{\mathrm{WCC}}^{\mathrm{phys}} = P$ or $NP_{\mathrm{WCC}}^{\mathrm{phys}} = NP$.

Theorem 5.2 (Verifier inclusion; correction of the empty-NP claim)**.** *In any model whose NP verifiers may ignore their witnesses and which subjects deciders and verifiers to the same resource rails,*

$$P_{\mathrm{WCC}}^{\mathrm{phys}} \subseteq NP_{\mathrm{WCC}}^{\mathrm{phys}}.$$

Proof. Given a polynomial-resource decider $D(x)$, define $V(x, w) = D(x)$ for a fixed empty witness (or ignore w). It has the same resource usage and correctness. Thus each physical-P language has a physical-NP verifier. □

Remark 5.3 (WIN-16)**.** *The original assertion $NP_{\mathrm{WCC}}^{\mathrm{phys}} = \varnothing$ contradicts the claimed nonempty physical-P class together with Theorem 5.2. No physical class separation follows from that assertion.*

5.3 A Correct Comparison Statement

Proposition 5.4 (No logical conflict between the two semantics)**.** *Suppose that the discrete classes equal classical P, NP, but physical admissibility imposes additional restrictions. It is logically possible that the physical classes are proper subsets of their respective discrete classes. No equality or inequality between physical P and NP follows from those facts alone.*

Proof. A class inclusion into a larger class does not determine whether two subclasses are equal. In particular the implication $P^{\mathrm{phys}} \subseteq P^{\mathrm{disc}}$ and $NP^{\mathrm{phys}} \subseteq NP^{\mathrm{disc}}$ is compatible with both $P^{\mathrm{phys}} = NP^{\mathrm{phys}}$ and a strict inclusion. □

5.4 On the Claimed Ill-Posedness

Remark 5.5 (WIN-17)**.** *Classical P versus NP is a precise question about languages and Turing machines. What can be* underspecified *is the expression "physically efficient computation" unless the physical machine, precision, allowed resources, and asymptotic family are specified. Failure of a physical equivalence theorem cannot make the classical mathematical question ill-posed.*

6 Locked Expanders and Physical Embedding Costs

A bounded-degree expander has an edge boundary of order the size of a balanced vertex subset. That is an $\Omega(n)$ combinatorial cut constraint on $O(n)$ vertices; it does not by itself prove that *one* encoded satisfying assignment must have $2^{\Omega(n)}$ interface area. Counting exponentially many assignments is different from physically storing all assignments at once.

Proposition 6.1 (Conditional interface-to-curvature lower bound)**.** *Suppose that for each admissible realization ψ of an instance F_n there is a geometric interface $\Gamma(\psi)$ such that*

$$\mathcal{H}^2(\Gamma(\psi)) \geq A_n \quad \textit{and} \quad C_{\Theta,\varepsilon}[\psi] \geq \kappa\, \mathcal{H}^2(\Gamma(\psi))$$

with a constant $\kappa > 0$ uniform in n, the input, and the encoding. Then every such realization satisfies $C_{\Theta,\varepsilon}[\psi] \geq \kappa A_n$.

Proof. Compose the two assumed inequalities. □

Remark 6.2 (WIN-18)**.** *A proved $A_n \geq 2^{\beta n}$, uniformly for every admissible implementation of a specific* decision problem*, would be a substantive new embedding theorem. Graph expansion alone does not establish it, and the original exponential-domain-wall theorem is therefore withdrawn as an unconditional conclusion.*

Proposition 6.3 (Verification of a supplied assignment)**.** *The language*

$$L_{\text{check}} = \{(F, \alpha) : \alpha \textit{ satisfies the 3-CNF formula } F\}$$

is in classical P.

Proof. Evaluate each literal of each clause under the supplied assignment and accept exactly when every clause contains a true literal. The runtime is polynomial in the encoded formula and assignment lengths. Restricting to uniformly constructible expander formulas does not make this verification task NP-hard. □

7 A Conditional Constructibility Barrier

Definition 7.1 (Per-step curvature increment)**.** *For a trajectory $\psi_{t+1} = F_n(\psi_t)$ set $\Delta C_{\Theta,\varepsilon}(t) = C_{\Theta,\varepsilon}[\psi_{t+1}] - C_{\Theta,\varepsilon}[\psi_t]$. A bound $|\Delta C_{\Theta,\varepsilon}(t)| \leq K(n)$ is a* separate hypothesis*; finite precision and local update radius alone do not imply a uniform constant K for all system sizes.*

Theorem 7.2 (Conditional telescoping capacity estimate)**.** *If $C_{\Theta,\varepsilon}[\psi_0] \leq p_0(n)$, $|\Delta C_{\Theta,\varepsilon}(t)| \leq K(n)$ for all $t < T(n)$, $T(n) \leq p_T(n)$, and p_0, K, p_T are polynomials, then*

$$C_{\Theta,\varepsilon}[\psi_{T(n)}] \leq p_0(n) + p_T(n)K(n) = \text{poly}(n).$$

Proof. Telescope the increments and use the triangle inequality:

$$C_{\Theta,\varepsilon}[\psi_T] = C_{\Theta,\varepsilon}[\psi_0] + \sum_{t<T} \Delta C_{\Theta,\varepsilon}(t) \leq p_0(n) + T(n)K(n).$$

□

Remark 7.3 (WIN-19)**.** *A high-curvature* witness *that violates the physical verifier rail is not evidence that its language belongs to physical NP. A separation requires one common language L with a polynomial-resource physical verifier for every admissible witness and a proof that no permitted polynomial-resource physical decider recognizes L. A lower bound only on one chosen witness representation is insufficient.*

8 Turing Simulability of Discrete WCC

The discrete simulation of Theorem 5.1 is an algebraic statement about locally updated finite symbols. With $N(n)$ cells and $T(n)$ global rounds, a sequential machine can evaluate the entire evolution in $T(n)\,\text{poly}(N(n))$ time. This does not convert an arbitrary physical PDE into the desired finite-state machine, nor does it charge curvature resources to the simulation. Such a cost-preserving realization must be established independently (see [15]).

9 Classical-to-Physical Implications

The extended Church–Turing thesis addresses simulation of physical computations in an abstract model with polynomial overhead. A one-directional simulation such as $P^{\mathrm{phys}} \subseteq P$ does not entail $P^{\mathrm{phys}} = P$, nor does it identify physical NP with classical NP. A physical equality claim would require the relevant simulation in both directions and preservation of witnesses, encodings, errors, and budgets.

Remark 9.1. *Even the hypothetical equality $P = NP$ would not, solely by definition, establish an equality between two separately defined physical classes. Conversely, unequal physical subclasses would not by themselves resolve classical P versus NP. No "if and only if ECT" criterion is proved or assumed here.*

10 Conclusion: What Is Established and What Remains Open

The corrected statements are: (i) under explicit uniformity, discrete WCC has the standard polynomial Turing simulation and verifier equivalence; (ii) positive storage or curvature costs require device-specific coercivity assumptions; (iii) H^2 control together with a positive regularization parameter gives a rigorous curvature upper bound; (iv) expander connectivity does not independently establish an exponential lower bound on the curvature of every decision procedure; and (v) Boolean influence cannot provide an exponential lower bound because it never exceeds n.

The proposed physical separation remains an open mathematical program. A viable theorem must name a language, specify its uniform physical encoding and resource rails, provide an admissible polynomial-resource physical verifier, and prove a lower bound against every admissible physical decider, not only against a particular geometric search protocol. Classical P versus NP remains mathematically well posed and unresolved by these constructions.

Formal verification status (November 2026). Lemma 4.2 and Theorem A.3 are kernel-checked as Lebesgue-integral inequalities on an arbitrary measure space; Proposition 2.2 is checked with the genuine Laplacian; Proposition A.1 is checked on the periodic cube with the stated constants; Lemma B.1, Theorem B.3 and Corollary B.4 are checked exactly. In the discrete model the Lean theorem `PVNP.Discrete.P_TM_eq_curvature_rail` shows that polynomial curvature rails exclude no polynomial-time language, and `PVNP.Discrete.curvature_not_representatio` shows that the curvature functional, unlike running time, depends on the chosen representation. For the discrete classes, Theorem 5.2 holds without its closure premise (`PVNP.Discrete.P_TM_subset_NP_TM` and `PVNP.Discrete.P_WCC_subset_NP_WCC`, via an explicit pair-parsing verifier machine). Proposition 6.3 is kernel-checked for a dense CNF encoding by an explicit verifier machine (`PVNP.Discrete.SATV.V_SA` which also gives CNF-SAT $\in NP$ in both discrete models (`PVNP.Discrete.SATV.SAT_dense_mem_NP_TM`). Theorem 5.1 is kernel-checked in dimension one for every fixed radius $r \geq 1$ (`PVNP.Discrete.P_WCCR_eq_P_TM`, `PVNP.Discrete.NP_WCCR_eq_NP_TM`); on $\mathbb{Z}^{1+d}$ only the inclusion of P_{TM} is (`PVNP.Discrete.P_TM_subset_P_WC`

Encoders and decoders. For the exact constant-amplitude solutions of the frozen PDE (`PVNP.WCTPlaneWave.plane_wave_solves`), an unrestricted encoder decides every language at constant mass, zero curvature energy and two final states (`PVNP.PhysicalAudit.unrestricted_encoder_triv` so does an unrestricted decoder after a fixed injective encoder (`unrestricted_decoder_trivializes`). With polynomial-time indicator encoders the constant-cost physical class is exactly P_{TM} (`PhysConst_polyEnc_e` and a physical lower bound for SAT over any encoder class containing them would prove $P_{\mathrm{TM}} \neq NP_{\mathrm{TM}}$ (`physical_SAT_lower_bound_is_classical`). A universal physical SAT lower bound is therefore at least as hard as the classical problem.

The flux inequality. The transport theorem $\int_0^T \|\Delta\psi\|_2^2 \, dt \geq \delta^4/(\Gamma^4 T^3)$ is kernel-checked from the instantaneous flux inequality $|J'(t)| \leq \Gamma\sqrt{D(t)}$ (`PVNP.TransportCalculus.curvature_time_lower_bound` for the continuum PDE the flux inequality itself is derived only analytically (integration by parts,

`pvnp/proofs/wct-pde-transport-bound.md`) and remains a premise of the Lean theorem. For the method-of-lines discretization on the discrete torus $\mathbb{Z}/N\mathbb{Z}$ it is *derived* in Lean: mass and Hamiltonian are exactly conserved (`PVNP.WCTLattice.mass_conserved`, `ham_conserved`), $\|Az\|^2 \le H + \frac{3\beta^2}{16\gamma} M$ uniformly in N (`bandpass_le_ham`), $J' = 2\,\mathrm{Im}\langle [A,w]z, Az\rangle$ (`hasDerivAt_flux`), and $|J'(t)| \le \Gamma\sqrt{\|Az(t)\|}$ with an explicit Γ fixed by the initial data (`lattice_flux_inequality`), giving the lattice curvature–time bound (`lattice_curvature_time_bound`).

Well-posedness. Global $H^2(\mathbb{R}^2)$ well-posedness, continuous dependence and conservation for the same equation are kernel-checked in the external library `rickyjreyes/math` (commit `30a5ddc`, `ReyesMath.Sobolev.global_wellposed`, `continuous_dependence`, `conservation`); the bridge `lean/bridge/MathBridge.lean` proves that its operator $(\Delta + k^2)^2$, nonlinearity and energy density coincide with the definitions used here ($\kappa = k^2$). These results are on the whole plane; the periodic and expanding-torus geometry of the physical model is not covered, and no uniform-in-L Poincaré constant exists there (`PVNP.SpectralGap.no_uniform_poincare`, sharp constant $(2\pi/L)^2$). Conditional statements remain conditional.

A Conditional Curvature–Capacity Analysis

A.1 Finite first-derivative energy is insufficient

Proposition A.1 (A high-frequency counterexample)**.** *On the periodic cube* $\Omega = (0, 2\pi)^3$*, let*

$$\psi_N(x) = 1 + \frac{1}{N}\sin(Nx_1), \qquad N \geq 2.$$

Then $\int_\Omega |\nabla\psi_N|^2 dx = 4\pi^3$ *and* $\int_\Omega |\Delta\psi_N|^2 dx = 4\pi^3 N^2$*. With* $\varepsilon = 1$ *in Definition 2.1,*

$$C_{\Theta,1}[\psi_N] \geq \frac{16\pi^3}{25} N^2 \longrightarrow \infty,$$

while the ordinary nonnegative mass, quartic, and gradient contributions of a fixed-coefficient first-derivative field-energy functional are uniformly bounded.

Proof. $\nabla\psi_N = (\cos(Nx_1), 0, 0)$ and $\Delta\psi_N = -N\sin(Nx_1)$, giving the stated integrals. Since $1/2 \leq \psi_N \leq 3/2$, the elementary inequality $\psi_N/(1 + \psi_N^2) \geq 2/5$ yields $|\Theta_1[\psi_N]| \geq (2/5)|\Delta\psi_N|$. Integrating gives the claimed bound. The amplitude stays between $1/2$ and $3/2$, so its L^2 and L^4 contributions remain bounded. □

Remark A.2 (WIN-14)**.** *Finite energy involving* $\|\nabla\psi\|_2^2$ *does not imply a uniform* H^2 *estimate, and generic finite action does not silently supply it. A positive* H^2 *regularity assumption or a proved PDE estimate is required.*

A.2 Locality and gravitational considerations

Lieb–Robinson bounds concern information propagation in suitable short-range quantum lattice systems; they do not imply that arbitrary PDE second derivatives grow at most polynomially in time. The original argument also identified divergence of a chosen normalized Laplacian functional with divergence of the stress–energy tensor without an established inequality between them. Einstein's equation by itself cannot close that missing step. The high-frequency family above is a counterexample to a universal fixed-energy bound on the chosen curvature functional.

Theorem A.3 (Corrected conditional capacity law)**.** *Assume a uniform family of WCT fields* $\psi_{n,x}(t) \in H^2(\Omega_n)$ *with a fixed* $\varepsilon > 0$ *and*

$$\sup_{|x|=n} \sup_{t \leq T(n)} \|\Delta\psi_{n,x}(t)\|_{L^2(\Omega_n)} \leq q(n)$$

for one polynomial q*. Then the regularized curvature cost obeys*

$$\sup_{|x|=n} \sup_{t \leq T(n)} C_{\Theta,\varepsilon}[\psi_{n,x}(t)] \leq \frac{q(n)^2}{4\varepsilon^2}.$$

Proof. Apply Lemma 4.2 uniformly over the family. □

Remark A.4 (WIN-15)**.** *This theorem is conditional on the missing uniform second-derivative estimate. Finite energy, spatial locality, and finite runtime alone do not provide the hypothesis. No universal QFT/GR/Lieb–Robinson polynomial curvature-capacity law is claimed.*

B Boolean Influence and Hypercube Boundary

For a Boolean function $f : \{0,1\}^n \to \{0,1\}$, let $x^{\oplus i}$ denote bit i flipped. Define

$$\mathrm{Inf}_i(f) = 2^{-n}\#\{x : f(x) \neq f(x^{\oplus i})\}, \qquad \mathrm{Var}(f) = \sum_{i=1}^{n} \mathrm{Inf}_i(f).$$

Define the *directed* boundary $B_{\to}(f) = \#\{(x,i) : f(x) \neq f(x^{\oplus i})\}$ and the *undirected* boundary $B_{\mathrm{und}}(f) = \frac{1}{2}B_{\to}(f)$.

Lemma B.1 (Corrected boundary–variation identities)**.** *For every Boolean f,*

$$B_{\to}(f) = 2^n \,\mathrm{Var}(f), \qquad B_{\mathrm{und}}(f) = 2^{n-1}\,\mathrm{Var}(f).$$

Proof. Count the $2^n \,\mathrm{Inf}_i(f)$ directed edges in coordinate i and sum over i; every undirected edge has two orientations. □

Remark B.2 (WIN-20)**.** *The original definition counted directed pairs while its lemma used the undirected normalization. Both formulas are now displayed separately.*

Theorem B.3 (Universal total-influence bound)**.** *For any Boolean function of n bits,*

$$0 \leq \mathrm{Var}(f) \leq n.$$

Consequently, it is impossible to have $\mathrm{Var}(F_n) = 2^{\Omega(n)}$ for any family of Boolean functions F_n with n variables.

Proof. Each $\mathrm{Inf}_i(f)$ is a probability in $[0,1]$. Summing n terms proves the upper bound. □

Corollary B.4 (Polynomial variation is vacuous as a complexity class)**.** *If P_{var} denotes Boolean language families with $\mathrm{Var}(f_n) \leq n^{O(1)}$ for all n, then P_{var} contains every Boolean language family. Thus the proposed exponential-variation exclusion of SAT and the resulting argument for $P \neq NP$ do not hold.*

Proof. Apply Theorem B.3 with exponent one. □

Remark B.5 (WIN-21)**.** *A large* unnormalized *cube boundary can be exponential even for simple functions; for example parity has $\mathrm{Var}(f) = n$ and $B_{\to}(f) = n2^n$. Neither an exponential directed boundary nor graph expansion by itself implies classical algorithmic hardness. The previous theorems claiming an exponential total influence and a strict subclass of NP have been replaced by these exact bounds.*

C Arrow of Time: Limits of the Claim

The thermodynamic arrow and the availability of time-reversing or closed-timelike-curve models are additional physical hypotheses, not logical characterizations of classical P and NP. A process computing a witness need not run the verifier backward: search algorithms operate forward in ordinary time. Entropy-reducing maps need not send information from the future to the past, and inverse dynamics need not efficiently solve SAT. Conversely, a generic assertion of retrocausality does not specify the special self-consistent fixed-point model considered in [7, 8].

Remark C.1. *No equivalence $P^{\mathrm{phys}} = NP^{\mathrm{phys}} \iff$ retrocausality, and no physical separation inferred merely from a forward thermodynamic arrow, is proved. This section is retained as a research direction rather than as a theorem.*

D Mathematical Revision Register

The following identifiers refer to the issue windows attached to this revised manuscript. "Corrected" means a valid replacement statement has been supplied; "conditional" means the original conclusion is not established and a precise additional hypothesis is specified.

WIN-12
Zero-curvature counterexample to the distinct-state lemma (Proposition 2.2); corrected.

WIN-13
Unsafe denominator $|\psi + \varepsilon| \geq \varepsilon$; regularization/gap distinction (Definition 2.1); corrected.

WIN-14
First-derivative energy does not control H^2; explicit counterexample (Proposition A.1); corrected.

WIN-15
Universal polynomial curvature growth from locality; replaced by conditional H^2 capacity (Theorem A.3).

WIN-16
Impossible assertion physical NP is empty; verifier inclusion (Theorem 5.2); corrected.

WIN-17
Classical mathematical ill-posedness; replaced by a distinction between formal and physical questions (Section 5); corrected.

WIN-18
Unsupported expander-to-exponential-interface implication; conditional interface bound (Proposition 6.1).

WIN-19
Inadmissible high-curvature NP witnesses; common physical verifier rail specified (Section 7).

WIN-20
Directed/undirected cube boundary factor of two; exact identities (Lemma B.1); corrected.

WIN-21
Exponential Boolean total influence; universal bound and vacuity of P_{var} (Theorem B.3); corrected.

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